Exercise for the Lecture
NLP with Deep Learning

TrustHLT

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Winter Semester 2026/27

Exercise Sheet 1

Exercise categories

The exercises are divided into two categories:

[Essential] These exercises cover concepts and skills that you are expected to understand and master.

[Training] These exercises provide more challenging problems for deepening and applying the essential concepts.

Classification Evaluation

Accuracy, precision, recall, and F1 are common measures for evaluating classification systems. In NLP, choosing an appropriate evaluation measure is particularly important because class distributions are often imbalanced.

Assume we build a simple Part-of-Speech classifier operating on the three tags `NN`, `VB`, and `ADJ`.

Exercise 1.1: Reading a Confusion Matrix

[Essential]

Consider the following confusion matrix:

		Predicted class		
		`NN`	`VB`	`ADJ`
True class	`NN`	25	5	1
	`VB`	2	15	12
	`ADJ`	1	6	0

For each class:

1. Identify TP, FP, and FN.
2. Compute precision.
3. Compute recall.
4. Compute F1.

Also compute the overall accuracy.

For class i:

$$
\begin{aligned}
P_i &= \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \\
R_i &= \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \\
F1_i &= \frac{2P_iR_i}{P_i + R_i}.
\end{aligned}
$$

What should happen to F1 when both precision and recall are zero?

Suggested solution

There are 67 instances in total and 40 correct predictions. Therefore,

$$\mathrm{Accuracy} = \frac{40}{67} \approx 0.597.$$

For `NN`:

$$TP = 25, \qquad FP = 3, \qquad FN = 6.$$

Thus,

$$
\begin{aligned}
P_{\mathrm{NN}} &= \frac{25}{28} \approx 0.893, \\
R_{\mathrm{NN}} &= \frac{25}{31} \approx 0.806, \\
F1_{\mathrm{NN}} &\approx 0.847.
\end{aligned}
$$

For `VB`:

$$TP = 15, \qquad FP = 11, \qquad FN = 14.$$

Thus,

$$
\begin{aligned}
P_{\mathrm{VB}} &= \frac{15}{26} \approx 0.577, \\
R_{\mathrm{VB}} &= \frac{15}{29} \approx 0.517, \\
F1_{\mathrm{VB}} &\approx 0.545.
\end{aligned}
$$

For `ADJ`:

$$TP = 0, \qquad FP = 13, \qquad FN = 7.$$

Therefore,

$$P_{\text{ADJ}} = 0, \qquad R_{\text{ADJ}} = 0.$$

The mathematical expression for F1 gives $0/0$ and is therefore undefined. In practical evaluation libraries, this case is commonly assigned an F1 score of 0 according to the library's zero-division convention.

Exercise 1.2: Macro and Micro Averaging

[Essential]

Using the confusion matrix from Exercise 1.1:

1. Compute macro precision.
2. Compute macro recall.
3. Compute macro-F1 by averaging the three per-class F1 scores.
4. Explain in words what macro averaging means.
5. What is micro-F1 for this single-label multiclass classification problem? How does it relate to accuracy?

Suggested solution

Using the values above:

$$P_{\text{macro}} = \frac{0.893 + 0.577 + 0}{3} \approx 0.490,$$

and

$$R_{\text{macro}} = \frac{0.806 + 0.517 + 0}{3} \approx 0.441.$$

Averaging the per-class F1 scores gives

$$F1_{\text{macro}} = \frac{0.847 + 0.545 + 0}{3} \approx 0.464.$$

Macro averaging assigns equal importance to every class, regardless of how many examples of that class occur in the dataset.

For single-label multiclass classification, each incorrect prediction creates one false positive and one false negative globally. Therefore, micro precision, micro recall, and micro-F1 are all equal to accuracy:

$$F1_{\text{micro}} = \frac{40}{67} \approx 0.597.$$

Exercise 1.3: Choosing an Evaluation Metric

[Essential]

For each scenario below, discuss whether accuracy, macro-F1, or both should be reported. Briefly justify your answer.

1. A sentiment dataset containing approximately the same number of positive and negative examples.
2. A toxicity classifier where only 2% of examples are toxic.
3. A POS tagger where common tags occur millions of times but some linguistically interesting tags are rare.

Suggested solution

1. Accuracy is informative because the classes are balanced, although reporting F1 can still be useful.
2. Accuracy alone is misleading. A classifier predicting "non-toxic" for every example would achieve 98% accuracy. Macro-F1 or class-specific precision and recall should therefore also be reported.
3. Accuracy reflects overall tagging performance, while macro-F1 reveals whether the model also performs well on rare tags. Reporting both gives complementary information.

Exercise 1.4: The Accuracy Trap

[Training]

A test dataset contains 990 examples of `classA` and 10 examples of `classB`.

Two classifiers are evaluated:

- `Model-One` predicts `classA` for every example.
- `Model-Two` predicts each class independently with probability 0.5.

For both systems:

1. Write down the expected confusion matrix.
2. Compute accuracy.
3. Compute precision, recall, and F1 for both classes.
4. Compute macro-F1.
5. Compare the systems. Which metric tells the most useful story about their behavior?

Suggested solution

For `Model-One`, the confusion matrix is

	A	B
A	990	0
B	10	0

and therefore

$$\text{Accuracy} = 0.99.$$

For class A,

$$P_A = \frac{990}{1000} = 0.99, \qquad R_A = 1,$$

and

$$F1_A \approx 0.995.$$

For class B, recall is zero and its F1 score is zero.
Thus,

$$F1_{\text{macro}} \approx \frac{0.995 + 0}{2} \approx 0.497.$$

For `Model-Two`, in expectation:

	A	B
A	495	495
B	5	5

so

$$\text{Accuracy} = 0.5.$$

For A,

$$P_A = \frac{495}{500} = 0.99, \qquad R_A = \frac{495}{990} = 0.5,$$

giving approximately

$$F1_A \approx 0.664.$$

For B,

$$P_B = \frac{5}{500} = 0.01, \qquad R_B = \frac{5}{10} = 0.5,$$

giving approximately

$$F1_B \approx 0.0196.$$

Hence,

$$F1_{\text{macro}} \approx 0.342.$$

Interestingly, macro-F1 still prefers `Model-One` even though `Model-One` completely ignores class B.

This demonstrates that no single metric fully describes model behavior. For strongly imbalanced tasks, per-class metrics are especially important.

Exercise 1.5: Implementing Evaluation

[Training]

Implement a confusion matrix from scratch in Python. You may use `numpy`, but do not use an evaluation library.

Your function should accept

```
y_true = [...]
y_pred = [...]
labels = [...]
```

and return a confusion matrix.

Then implement:

1. per-class precision,
2. per-class recall,
3. per-class F1,
4. macro-F1,
5. micro-F1.

Consider explicitly what your implementation should do when a denominator is zero.

Exercise 1.6: Comparing Against scikit-learn

[Training]

Use the corresponding metrics from `scikit-learn` to verify your implementation:

`https://scikit-learn.org/stable/modules/model_evaluation.html`

Experiment with the following values of the `average` argument:

- `micro`,
- `macro`,
- `weighted`.

Construct a small imbalanced dataset for which macro-F1 and weighted-F1 differ substantially.

Explain why they differ.

Text Generation Evaluation

Unlike classification, text generation often has no single correct output. A generated sentence may be perfectly valid even if it differs from a reference sentence. This makes automatic evaluation considerably harder.

Exercise 1.7: Reference-Based Evaluation

[Essential]

Consider the reference:

> *The cat is sitting on the mat.*

and two system outputs:

> **A:** *The cat is sitting on the mat.*
>
> **B:** *A cat sits on the mat.*

Without calculating exact scores:

1. Which output should obtain the higher BLEU score?
2. Which should obtain the higher ROUGE score?
3. Is output B necessarily worse than output A?
4. What does this example tell us about reference-based evaluation?

Suggested solution

Output A should obtain the highest score under both metrics because it matches the reference exactly.

Output B is semantically very similar, but differs lexically and syntactically. It will therefore generally receive a lower reference-based score.

This does not imply that B is worse. BLEU and ROUGE measure forms of overlap with reference texts rather than semantic correctness directly.

Exercise 1.8: BLEU versus ROUGE

[Essential]

Answer briefly:

1. What does BLEU primarily measure?
2. What does ROUGE primarily measure?
3. Why was BLEU historically popular for machine translation?
4. Why was ROUGE historically popular for summarization?
5. Can either metric directly determine whether a generated statement is factually correct?

Suggested solution

BLEU primarily measures n-gram precision between a generated text and one or more references,

together with a brevity penalty.

ROUGE is a family of overlap measures commonly emphasizing how much of the reference content is recovered by the generated text.

BLEU became popular in machine translation because good translations often share substantial phrase-level overlap with reference translations.

ROUGE became popular in summarization because recovering important reference content is central to many summarization settings.

Neither metric directly evaluates factual correctness.

Exercise 1.9: When Overlap Metrics Fail

[Training]

Consider the reference:

> *Berlin is the capital of Germany.*

and the following outputs:

> **A:** *Berlin is the capital of Germany.*
>
> **B:** *Germany's capital is Berlin.*
>
> **C:** *Berlin is not the capital of Germany.*

Discuss:

1. Which outputs have high lexical overlap with the reference?
2. Which outputs are semantically correct?
3. Why might an overlap-based metric give an undesirable ranking?
4. What property of language makes generation evaluation harder than ordinary classification evaluation?

Suggested solution

A is an exact match and is correct.

B has lower lexical overlap but preserves the meaning and is correct.

C has extremely high lexical overlap but reverses the meaning by inserting "not" and is therefore incorrect.

An overlap metric may consequently assign C a higher score than B despite B being the correct output.

Natural language allows many surface forms to express approximately the same meaning, while very small surface changes can radically alter meaning.

Exercise 1.10: Metric Design

[Training]

Suppose you are evaluating a system that generates answers to factual questions.

Would BLEU or ROUGE alone be sufficient?

Propose at least three properties that an ideal evaluation method should consider.

For each property, explain why lexical overlap with a reference may fail to measure it.

LLMs as Judges

Modern NLP systems are increasingly evaluated using another language model as an evaluator, commonly called an *LLM-as-a-Judge*.

Instead of comparing outputs only through lexical overlap, a judge model receives one or more generated answers and evaluates them according to specified criteria.

Exercise 1.11: What Is an LLM-as-a-Judge?

[Essential]

Suppose we ask:

> *Why do leaves appear green?*

A system generates:

> *Leaves appear green because chlorophyll absorbs much of the red and blue light while reflecting more green light.*

Instead of computing BLEU or ROUGE, we give the question and answer to a powerful language model and ask it to assign a score from 1 to 5 for correctness.

Answer the following:

1. What information can the LLM judge use that BLEU cannot?
2. What is one major advantage of this evaluation method?
3. What is one major risk?
4. Does using a stronger LLM automatically make the evaluation objective? Explain.

Suggested solution

An LLM judge can reason about semantic content, factual claims, relevance, coherence, and instructions rather than relying only on lexical overlap.

A major advantage is that it can evaluate open-ended outputs for which many different answers may be valid.

A major risk is that the judge itself may make mistakes or exhibit systematic biases.

A stronger model does not make the evaluation objective. The resulting scores still depend on the judge model, prompt, evaluation criteria, and possibly properties of the candidate response unrelated to its actual quality.

Exercise 1.12: Pointwise and Pairwise Evaluation

[Essential]

Consider two evaluation procedures.

Procedure A

> Rate the following answer from 1 to 5 for factual correctness.

Procedure B

> Given answers A and B, determine which answer is more factually correct. Respond with A, B, or Tie.

1. Which procedure is pointwise evaluation?
2. Which procedure is pairwise evaluation?
3. Give one advantage of pairwise evaluation.
4. Give one disadvantage of pairwise evaluation.

Suggested solution

Procedure A is pointwise evaluation.

Procedure B is pairwise evaluation.

Pairwise comparison may be easier because the judge only needs to decide which of two responses better satisfies a criterion rather than map quality onto an absolute numerical scale.

A disadvantage is that evaluating many systems may require many pairwise comparisons.

Pairwise evaluation can also suffer from biases related to the ordering of the candidate responses.

Exercise 1.13: Designing a Judge Prompt

[Essential]

Suppose an LLM should judge generated summaries.

Which of the following prompts is preferable?

Prompt A

> Is this summary good? Give it a score.

Prompt B

> Evaluate the summary on factual consistency, coverage of the important information, and conciseness. Score each criterion from 1 to 5. Base your evaluation only on the provided source document.

Explain your choice.

What additional information or instructions could make the evaluation more reproducible?

Suggested solution

Prompt B is preferable because it defines explicit evaluation dimensions, a scoring range, and the evidence on which the evaluation should be based.

Reproducibility can be improved by defining the meaning of each score, providing examples, specifying how conflicting criteria should be handled, fixing the judge model and decoding configuration, and using the same prompt format for all evaluated systems.

Exercise 1.14: Position Bias

[Training]

An LLM judge compares two answers, A and B.

When prompted with

> **Answer A:** X
>
> **Answer B:** Y

the judge chooses A.

After swapping the order,

> **Answer A:** Y
>
> **Answer B:** X

the judge again chooses A.

1. What problem does this experiment reveal?
2. How could you modify the evaluation procedure to detect this systematically?
3. How could you aggregate judgments from both orders?
4. What should happen if the judge gives contradictory decisions?

Suggested solution

The behavior suggests position bias: the judge prefers the response appearing in a particular position independently of its content.

Each pair can be evaluated twice with reversed ordering.

A robust procedure may accept a winner only when the decisions are consistent after accounting for the swap.

Inconsistent comparisons can be treated as ties, marked as uncertain, or sent for additional evaluation.

The important point is that the evaluation protocol should explicitly define how such inconsistencies are handled.

Exercise 1.15: Judge Biases

[Training]

For each situation, identify a potential problem with LLM-based evaluation.

1. Two answers are equally correct, but one is three times longer and receives a higher score.
2. A model is used to judge outputs generated by the same model family and consistently prefers them.
3. A stylistically polished answer containing a subtle factual error beats a terse but correct answer.
4. Changing "Rate the answer" to "Be a strict evaluator and rate the answer" substantially changes the scores.

For each problem, suggest one experiment that could reveal or quantify the effect.

Suggested solution

Potential issues include:

1. verbosity bias,
2. self-preference or model-family bias,
3. style bias or failure to detect factual errors,
4. sensitivity to the judge prompt.

Possible experiments include controlling response length, using judges from different model families, constructing adversarial answers with controlled factual errors, and evaluating the same outputs under multiple semantically similar judge prompts.

Exercise 1.16: Can We Trust the Judge?

[Training]

You evaluate 500 generated answers using an LLM judge.

To test the judge, you also ask human annotators to evaluate a subset of 100 answers. The LLM agrees with the human majority on 78 of them.

Discuss the following:

1. Is 78% agreement enough to conclude that the judge is reliable?
2. What additional information would you want before trusting the judge?
3. Why should human–human agreement also be measured?

4. Would high agreement guarantee that the judge is unbiased?

Suggested solution

The agreement number alone is insufficient.

We would also want to know the class or score distribution, the type and severity of disagreements, agreement under different judge prompts, the quality and agreement of the human annotators, and whether disagreements are systematic.

Human–human agreement provides an estimate of how well-defined the task itself is. If humans frequently disagree, expecting perfect agreement from an automated judge may be unrealistic.

High aggregate agreement also does not guarantee absence of bias. The judge may agree overall while systematically failing on particular answer types, topics, writing styles, or model families.

Exercise 1.17: Design an Evaluation Pipeline

[Training]

You have three LLMs that answer open-ended factual questions:

$$M_A, \qquad M_B, \qquad M_C.$$

You have 1,000 questions but no unique reference answer for each question.

Design an LLM-as-a-Judge evaluation protocol.

Your design should specify:

1. what information is shown to the judge,
2. whether you use pointwise or pairwise evaluation,
3. the evaluation criteria,
4. how you reduce position or ordering effects,
5. whether you use one judge or multiple judges,
6. how you validate the judge against humans,
7. how you aggregate the final results.

There is no single correct solution. Your goal is to design an evaluation whose assumptions, limitations, and potential biases are explicit.

Suggested solution

One possible design is pairwise evaluation.

For each question, responses from two systems are shown together with the original question and, when available, trusted evidence.

The judge evaluates factual correctness and relevance according to an explicit rubric.

Every pair is evaluated in both orders to detect position effects. Multiple judge models from different

model families may be used to reduce dependence on a single evaluator.

A randomly sampled subset should additionally be evaluated by human annotators. Human–human and human–judge agreement should be reported.

Pairwise wins can be aggregated into win rates or another explicitly defined ranking procedure. Ties and inconsistent swapped-order judgments should be retained rather than silently converted into wins.

Most importantly, the resulting ranking should be interpreted as a result under a particular evaluation protocol, not as an objective measurement of model quality.